

Literature Review: Multi-Crop Plant Leaf Disease Detection Using Deep Learning and Explainable AI

Introduction

Automated plant leaf disease detection using deep learning has matured rapidly over the past several years, with recent work converging on two dominant threads: (1) architectural hybridization — combining convolutional neural networks (CNNs) with transformer-based global attention to jointly capture local lesion texture and global leaf context, and (2) explainable AI (XAI) integration — using techniques like Grad-CAM, Grad-CAM++, LRP, and LIME to render model decisions interpretable for agricultural stakeholders. The five studies reviewed below span this landscape, from a heavyweight 92M-parameter hybrid ResNet-Transformer model to a lightweight 12M-parameter edge-deployable network, and from single-XAI-method qualitative visualization to a six-method XAI fusion approach. Read together, they reveal a field that has largely solved single-dataset, benchmark-level classification accuracy (consistently 95–99%+) but has not yet solved the harder problems of cross-dataset robustness under real field conditions, quantitatively validated interpretability, and genuinely multi-crop generalization — gaps that motivate the present project's three-crop, fifteen-class scope.

Hybrid CNN-Transformer Architectures

Two of the reviewed papers pursue essentially the same architectural strategy — fusing a CNN backbone with a Vision Transformer encoder — but at different scales and with different crop scopes. Srinivasan et al. (2026) propose ResViT-152, fusing a ResNet152V2 backbone (yielding a $7 \times 7 \times 2048$ feature map) with a six-block, eight-head ViT encoder over CBAM-refined features, evaluated across corn, tomato, and potato on two large Kaggle datasets (39,203 and 65,565 images, 18 classes total). Critically, this paper is one of the few in the set to run genuine cross-dataset evaluation (train on one dataset, test on the other), reporting a modest but real accuracy drop from ~99% intra-dataset to ~95–96% cross-dataset — a rare and useful signal, since most competing papers report only within-dataset performance. A second paper (banana-cherry-tomato, unnamed lead title in the review notes) proposes a more modest Hybrid ConvNet-ViT (~32–35M parameters) evaluated via 5-fold cross-validation on a single Kaggle dataset (bct-image-dataset, 9 classes, notably imbalanced — a 34× spread between the smallest and largest class). Its accuracy gain over a standalone ViT baseline is small (99.29% vs. 98.92%), and the paper is flagged as having credibility issues: a Related Work table apparently copy-pasted from an unrelated drowsiness-detection manuscript, and confusion matrices explicitly described as "simulated" from reported accuracy figures rather than raw empirical output. Both papers illustrate the same trade-off: hybrid CNN-Transformer fusion reliably buys a small accuracy edge over pure CNN or pure ViT baselines, but at real computational cost (92M and ~33M parameters respectively), and neither paper tests true field conditions or resource-constrained deployment.

Lightweight and Edge-Oriented Models

In contrast, OptiNet-B3 (Naveen & Ajitha, 2025) explicitly targets deployability. Built on an EfficientNet-B3 backbone augmented with Mish activation, CBAM attention, Group Normalization, and knowledge distillation from a larger teacher model, it achieves 98.12% (fruit) and 99.23% (leaf) accuracy across apple, banana, and orange datasets — while using only 12M parameters, 1.8B FLOPs, and 8.2ms inference time, roughly an order of magnitude lighter than the hybrid architectures above. This paper is a useful counterpoint in the reviewed set: it demonstrates that near-SOTA accuracy does not require transformer-scale parameter counts, but it does so at the cost of scope — only three crops, and no cross-dataset generalization testing, since both its fruit and leaf datasets are curated and evaluated in isolation.

Capsule Networks and Multi-Method XAI

CapsNet-XAI (paywalled; only abstract/metadata accessible) takes a distinct architectural route, using capsule networks with dynamic routing-by-agreement to preserve part-whole spatial relationships, paired with six separate XAI techniques (LRP, Grad-CAM, LIME, Integrated Gradients, SmoothGrad, Guided Backpropagation) rather than the single Grad-CAM/Grad-CAM++ approach used elsewhere in the set. It reports strong accuracy (98–99.5%) across four heterogeneous public datasets (PlantVillage1, PlantDoc, Mango, PlantVillage), and its inclusion of PlantDoc — a genuinely field-collected, cluttered-background dataset — makes it the only paper in the set testing on non-curated imagery, though full methodological detail is inaccessible behind the paywall. Its six-method XAI fusion is notably more rigorous in *breadth* than the single-technique Grad-CAM++ visualizations used in the hybrid-architecture papers, though even here, quantitative agreement between the six explanation methods is not confirmed from the abstract alone.

Convergent Themes and Research Gaps

Three patterns recur across all five papers and directly inform this project's positioning:

XAI remains qualitative, not quantitative. Every paper in the set that discusses explainability (Srinivasan et al.'s Grad-CAM++, OptiNet-B3's Grad-CAM/Grad-CAM++) reports a small set of example heatmaps rather than a systematic, quantitatively validated interpretability metric — e.g., IoU against expert-annotated lesion masks. Srinivasan et al. explicitly defer this to future work despite prominently featuring "explainable AI" in the title, and report a 93.5% "XAI interpretability score" that is not rigorously grounded. This is arguably the field's single largest credibility gap, and one this project's roadmap already addresses by planning quantitative Grad-CAM/Grad-CAM++ validation rather than illustrative-only heatmaps.

Curated data dominates; field conditions are rare. With the partial exception of CapsNet-XAI's use of PlantDoc, every dataset in this set is Kaggle-sourced and either lab-controlled or only semi-field (variable lighting/background but still single-leaf, single-disease framing). None address multi-disease co-occurrence on a single leaf, occlusion, or true in-field clutter at scale. This matches the project's own dataset characteristics (three crops, 15 classes, ~29,000 images) and underscores that cross-dataset or field-condition validation — not just within-dataset accuracy — is where genuine contribution is still possible.

Reproducibility is uniformly poor. No paper in the reviewed set releases code or trained weights; all five either omit a data-availability statement entirely or point to "available from the corresponding author on reasonable request" —

inconsistent given that the underlying datasets are already public on Kaggle. This is a field-wide weakness worth naming explicitly in the review's limitations discussion, and it's also a point of differentiation this project can address by planning to release its data pipeline and splitting methodology.

A secondary but important methodological gap, not discussed in any of the five papers, is **data leakage from pre-augmented images**: several of the suspiciously high (95–99%+) cross-dataset and within-dataset accuracies reported across this literature are plausible candidates for leakage if augmented variants of the same base image were allowed to span train/test splits — precisely the issue already identified and corrected in this project's own Papaya subset via group-aware `GroupShuffleSplit`. None of the reviewed papers mention group-aware splitting, which suggests this may be an under-recognized confound behind some of the field's near-ceiling accuracy claims.

Synthesis Table

Paper	Crops	Params	Architecture	Cross-Dataset Test	XAI Method	Best Accuracy
Srinivasan et al. (2026), ResViT-152	Corn, tomato, potato	~92M	ResNet152V2 + ViT + CBAM	Yes (bidirectional)	Grad-CAM++ (qualitative)	99.12% intra / 96.27% cross
Hybrid ConvNet-ViT (banana/cherry/tomato)	Banana, cherry, tomato	~33M	Conv stem + ViT encoder	No (5-fold CV only)	Grad-CAM (qualitative)	99.29%
CapsNet-XAI	Multi (4 datasets incl. PlantDoc)	Not stated	Capsule network, dynamic routing	Partial (4 heterogeneous datasets)	6-method fusion (LRP, Grad-CAM, LIME, IG, SmoothGrad, GBP)	99.46% (best dataset)
OptiNet-B3 (2025)	Apple, banana, orange	12M	EfficientNet-B3 + CBAM + KD	No	Grad-CAM/Grad-CAM++	99.23%

Positioning of the Current Project

Relative to this literature, the project's current trajectory — a clean, group-aware-split single-backbone baseline (EfficientNetB0/B3 or ResNet50) with macro-F1 evaluation before pursuing hybrid complexity, followed by quantitatively validated Grad-CAM/Grad-CAM++ explainability — directly targets the two weakest points identified across all five papers: unvalidated XAI claims and unaddressed data-leakage risk. The planned deferral of hybrid CNN-Transformer architecture until baseline integrity is established is well justified by this review, since none of the reviewed hybrid models' accuracy gains over simpler baselines (e.g., 99.29% vs. 98.92% in the banana-cherry-tomato paper) are large enough to justify added complexity without first confirming the baseline itself is leakage-free.